

## ROLE AND OBJECTIVE:

You are a senior biomedical researcher specialized in bioelectromagnetic and oncology. Your task is to perform a comprehensive analysis of the provided dataset (covering 49 scientific papers) attached to this prompt. The objective is to assess if specific electromagnetic field (EMF) configurations can serve as a non-thermal antitumor monotherapy, selectively targeting cancer cells ( $Is\_Cancer=1$ ) while sparing healthy ones ( $Is\_Cancer=0$ ). The goal is to identify hidden correlations and mechanistic patterns to formulate a novel experimental hypothesis for a future project.

## DATABASE STRUCTURE:

Each row of the table represents a unique experimental condition (combination of cell line, field type, frequency, intensity, exposure duration, observation time, etc.). Below is a complete description of all columns in the table.

### Metadata columns

Paper\_ID: paper number.

Year: publication year of the study.

Rank: journal ranking (SCImagojr).

Journal: name of the journal where the paper was published.

Study\_Type: in vitro study.

### Cell line parameters

Cell\_Line: cell lines used in the study. One row in the table for each cell line analyzed. If there are different field exposure conditions for each cell line, there is the same number of rows for each combination of parameters described in the study.

Culture\_Type: indicates whether the cell lines are adherent to the plate or are spheroids.

Is\_Cancer: "1" if the cell line is cancerous; "0" if it represents non-cancerous cells.

Type: type of cell line.

Origin: origin of the cell line, whether human or animal.

### Exposure parameter columns

Field\_Type: type of electromagnetic field used in the study.

Wave\_Type: field waveform used in the study.

AC\_Freq\_Value: frequency of the AC component (if used) used in the study.

AC\_Freq\_Unit: frequency unit.

Freq\_Rationale: two or three words on why that specific frequency was chosen.

Freq\_Rationale\_Detail: more detailed explanation of why that frequency was chosen.

Max\_Intensity\_AC\_Value: AC maximum intensity of the magnetic field.

Max\_Intensity\_AC\_Unit: unit of measurement for the AC component.

Intensity\_DC\_Value: DC intensity of the magnetic field.

Intensity\_DC\_Unit: unit of measurement for the DC component.

Electric\_Field\_Intensity\_Value: electric field intensity value.

Electric\_Field\_Intensity\_Unit: unit of measurement for the electric field intensity.

Power\_Density\_Value: power density value.

Power\_Density\_Unit: unit of measurement for power density.

SAR\_(W/kg): Specific Absorption Rate in W/kg.

Intensity\_Rationale: two or three words on why that specific intensity was chosen.

Intensity\_Rationale\_Detail: more detailed explanation of why that intensity was chosen.

Pulse\_Duration\_Value: pulse duration value.

Pulse\_Duration\_Unit: unit of measurement for pulse duration.

Pulse\_Repetition\_Rate\_Value: pulse repetition rate value.

Pulse\_Repetition\_Rate\_Unit: unit of measurement for pulse repetition rate.

Pulse\_Repetition\_Rate\_Notes: additional notes on pulse repetition rate.

Pulse\_Repetition\_Rate\_Detail: detailed explanation of the pulse repetition rate.

Number\_of\_Pulses: number of pulses used.

Field\_Notes: additional information regarding the field.

Device: device used to generate the field.

Thermal\_Mechanism: "0" if the study reported no thermal effects on the cells. "1" if the study reported thermal effects on the cells.

Control\_Group: how the control group not exposed to the field is treated.

### Cell fate outcome columns

For the cell fate endpoint Viability, Proliferation, Apoptosis, Necrosis, Senescence, Motility, the following columns exist (where [Name] is the outcome name):

[Name]\_Assay: assay used to measure the specific block [Name].

[Name]\_Exposure\_Duration\_h: exposure time, expressed in hours, during which the cell lines were subjected to the field to test the specific outcome [Name].

[Name]\_Exposure\_Duration\_h\_Notes: notes about the field exposure time for the specific outcome [Name].

[Name]\_Observation\_Time\_h: how much time after the end of the field exposure, expressed in hours, the results for this specific outcome [Name] are observed.

[Name]\_Exposed\_%\_vs\_Control: percentage of cells measured for the outcome [Name] after field exposure, expressed relative to the control group (set to 100%).

[Name]\_Direction: "-1" if there was a significant reduction in the value reported in the previous column; "1" if there was a significant increase; "0" if there was no significant increase or reduction. Significance is reported within the paper.

[Name]\_Notes: comments regarding block [Name] or the results obtained within block [Name] mentioned in the paper.

In the cell fate outcomes, there is also the Ferroptosis outcome, which has the following columns:

Ferroptosis\_Flag: "1" if the outcome was observed, "0" if was not observed even if measured.

Ferroptosis\_Notes: comment based on the information provided in the paper.

### Mechanistic outcome columns

ROS and Antioxidant follow the same column structure as cell fate outcomes.

Calcium\_Flag: "1" if involvement of Calcium channels was observed, "0" if involvement was not observed even if measured.

Calcium\_Notes: comment based on the information provided in the paper.

Cellular\_Morphology\_Flag: "1" if changes in cell morphology were observed, "0" if no changes were observed even if sought.

Cellular\_Morphology\_Observation\_Method: method used to observe the morphology of the cells involved in the study.

Cellular\_Morphology\_Exposure\_Duration\_h: exposure time, expressed in hours, during which the cell lines were subjected to the field to observe their morphology.

Cellular\_Morphology\_Observation\_Time\_h: how much time after the end of field exposure, expressed in hours, the cell morphology is observed.

Cellular\_Morphology\_Notes: what happened to the morphology of the cells.

Membrane\_Potential\_Assay: assay used to measure mitochondrial membrane potential.

Membrane\_Potential\_Target: structure whose membrane potential is measured.

Membrane\_Potential\_Exposed\_%\_vs\_Control: value of the mitochondrial membrane potential of exposed cells expressed relative to the control group (set to 100%).

Membrane\_Potential\_Direction: "-1" for a significant reduction; "1" for a significant increase; "0" for no significant change.

Membrane\_Potential\_Notes: notes on mitochondrial membrane potential reported in the paper.

Metabolic\_Reprogramming\_Flag: "1" if metabolic reprogramming was observed, "0" if not observed.

Metabolic\_Reprogramming\_Notes: comments stated in the paper regarding metabolic reprogramming.

Differentiation\_Flag: "1" if differentiation of the cells is investigated, "0" if investigated but no effects observed.

Differentiation\_Notes: information cited in the paper regarding differentiation.

Cytoskeleton\_Flag: "1" if effects on cytoskeleton are quantitatively measured, "0" if measured but no effects observed.

Cytoskeleton\_Target: cytoskeleton structure that was affected or discussed.

Cytoskeleton\_Notes: information cited in the paper regarding cytoskeleton.

Cell\_Cycle\_Phase: phases where cell cycle changes were observed/measured, separated by semicolons.

Cell\_Cycle\_Exposure\_Duration\_h: exposure time in hours for cell cycle observation.

Cell\_Cycle\_Observation\_Time\_h: hours after exposure when cell cycle changes are measured.

Cell\_Cycle\_Direction: "1" for significant increase; "-1" for significant reduction; "0" for no significant change.

Follows phase order, separated by semicolons.

Cell\_Cycle\_Notes: information on cell cycle reported in the paper.

### Molecular outcome columns

For the Genomic and Proteomic analysis, the following columns exist (where [Name] is the genomic or proteomic outcome):

[Name]\_Assay: assay used to study genes/proteins involved.

[Name]\_Target\_Family: family to which the studied genes/proteins belong.

[Name]\_List: list of genes/proteins studied, separated by semicolons.

[Name]\_Direction: "1" for significant increase; "-1" for significant reduction; "0" for no significant change.

Follows list order, separated by semicolons.

[Name]\_Detail: for each gene/protein, the name followed by a colon and a brief explanation of its role, then a period, proceeding to the next.

[Name]\_Notes: consequences of the change or lack of change in observed genes/proteins.

#### ANALYSIS INSTRUCTIONS:

Before producing any content, internalize the following principle: this analysis must reproduce the reasoning depth of a domain expert performing a literature-based analysis. You are not summarizing individual papers; you are synthesizing evidence between papers to identify patterns, contradictions, dose–response trends, mechanistic connections, and selectivity windows that no single study reveals by its own.

#### Step 1 – Preliminary filtering and grouping

Before any quantitative analysis, discard every row in which Thermal\_Mechanism = 1. From this point onward, only non-thermal conditions are considered. Report the number of rows excluded and the number retained. Next, partition the retained dataset into internally coherent subgroups based on the Field\_Type column. The authors of the primary studies have used heterogeneous and sometimes overlapping naming conventions for the same physical category of field, so you must not take Field\_Type values at face value. Inspect the distinct values present in the column, interpret each of them in terms of the underlying physics (frequency regime, presence or absence of a static component, pulsed vs continuous nature, electric vs magnetic dominance, etc.) and group them into a limited number of macrocategories that are large enough to support a subgroup–level analysis and homogeneous enough to allow meaningful comparisons within each other. Do not impose a predefined list: the grouping must emerge from what is present in the dataset. For each macrocategory you define, state explicitly which original Field\_Type labels have been merged into it and justify the grouping in two or three sentences. Before entering the subgroup–level analyses, produce a short gap–identification section: for each macrocategory, report how many experimental conditions and how many distinct papers (Paper\_ID) contribute to it, and show which biological outcomes and mechanisms have been investigated within each macrocategory and with what coverage. This serves to flag the macrocategories whose evidence base is too thin to support firm conclusions and the outcome–field combinations that remain unexplored.

#### Step 2 – Subgroup–level analysis

For each macrocategory you have defined, produce a self–contained analytical module. The modules must all follow the same internal logic:

1. Subgroup overview. State the number of experimental conditions, the number of contributing papers, the cancer–to–healthy ratio of the tested cell lines, and the per–study ratio of cancer vs healthy lines. Comment on whether the subgroup is dominated by a single tumor type or spans a broader landscape, and whether enough studies include both cancer and healthy lines to allow a direct selectivity assessment.
2. Exposure parameters. Describe the parameter space explored within the subgroup. Use a lookup table of the distinct experimental conditions (one row per unique combination of parameters). Report the distribution of the values tested and the rationales behind those choices.
3. Cell fate outcomes. Analyze the seven cell fate outcomes (viability, proliferation, apoptosis, necrosis, ferroptosis, senescence, motility). For each outcome: quantify how many conditions measured it, broken down by cancer vs healthy cells; describe the distribution of directions; identify the conditions that produced the strongest effects on cancer cells (reporting the exact parameters, the cell line, the exposure and observation times, and the %\_Exposed\_vs\_Control value); identify conditions that produced counterproductive effects on cancer cells; and, whenever both cancer and healthy lines were tested under matched conditions, explicitly assess selectivity by comparing the directions and magnitudes of the effect on the two cell populations, flagging conditions with good selectivity, negligible selectivity, and negative selectivity (i.e. damage to healthy cells). Where the data allow, comment on dose–response and time–response trends.

4. Mechanistic outcomes. Analyse the mechanistic outcomes in the following order: ROS production, antioxidant response, membrane potential, calcium signalling, cellular morphology and cytoskeleton remodelling, metabolic reprogramming, differentiation, cell cycle. For each mechanism: quantify the conditions that measured it, describe the observed effects separately for cancer and healthy cells, and trace the mechanistic flow from the mechanism to the downstream cell fate outcomes measured under the same conditions. When a mechanism is only “Mentioned”, state this explicitly and treat the evidence as qualitative.
5. Molecular outcomes. Analyse genomic and proteomic columns. Identify the gene and protein families most frequently investigated within the subgroup, the specific targets that recur across independent studies, and, for each recurrent target, if the direction of change is consistent across conditions or varies. Highlight cases in which the same target moves in opposite directions in cancer vs healthy cells under the same exposure.
6. Mechanism of action and connections within the subgroup. Synthesize the findings into a coherent proposed mechanism of action for the subgroup.
7. Limitations of the subgroup evidence base. Explicitly state the limitations internal to this subgroup.

### Step 3 – Cross-subgroup synthesis

After all subgroup modules are complete, produce a comparative section across macrocategories. Identify shared mechanistic signatures that appear across multiple subgroups and mechanisms that are specific to individual subgroups.

### Step 4 – Global limitations and conclusions

Close with a global limitations section (here address limitations of the dataset as a whole) and a conclusions section that synthesizes the therapeutic potential of non-thermal electromagnetic fields as cancer monotherapy, based strictly on the evidence analysed.

#### ANALYTICAL STANDARDS:

Be quantitative. Every claim about effects must be supported by values of %\_Exposed\_vs\_Control, exposure parameters, exposure and observation times, and cell line names. Vague statements of the form “the field reduces viability” are unacceptable unless accompanied by the specific conditions and numbers.

Do not invent data. If a field, a cell line or a mechanism was not tested under a given condition, state that it was not tested. If the source paper only mentions a mechanism without measuring it, use the qualitative \_Notes columns and make the qualitative nature of the evidence explicit.

When useful, recommend visualizations: numbered figures and tables are encouraged and must be referenced in the text.

#### OUTPUT FORMAT:

Produce the full analysis as a single report in a downloadable .docx document. The document must require no further editing or rearrangement.